

FeatureSAGE: Stable Attribute Group Editing With Integrated StyleFeatureEditor Encoder for Enhanced Few-shot Image Generation

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Abstract: *Few-shot image generation remains difficult because limited examples can cause mode collapse, reduced image quality, and unstable semantic edits. Stable Attribute Group Editing (SAGE) improves latent attribute editing for this setting, but its pixel2style2pixel (pSp) inversion path can lose local details and produce less reliable latent codes. This study proposes FeatureSAGE, a SAGE-based framework that integrates the StyleFeatureEditor Inverter and Feature Editor to represent images with both W^+ latent codes and intermediate feature tensors. The pretrained StyleGAN2 generator is kept fixed while feature-level editing is added to improve reconstruction detail and class-consistent variation. In one-shot experiments on FUNIT Animal Faces and 102 Flowers at 1024×1024 resolution, FeatureSAGE achieves FID scores of 45.60 and 49.60, outperforming SAGE and AGE while maintaining comparable LPIPS on Flowers and Animal Faces, respectively. These results indicate that joint latent-feature inversion and editing improve few-shot attribute group editing without removing useful perceptual variation.*

Keywords: Few-shot image generation; GAN inversion; StyleGAN2; StyleFeatureEditor; attribute group editing

1. INTRODUCTION

Generative Adversarial Networks (GANs) learn an adversarial game between a generator and discriminator and have become a standard foundation for high-fidelity synthesis [1]. StyleGAN and StyleGAN2 further improved image quality by mapping an input latent code into an intermediate style space and modulating synthesis layers, making semantic manipulation more practical than in the original latent space [2, 3]. These properties are useful for few-shot image generation, where a model must synthesize plausible samples for a novel category from very limited references rather than retraining a generator from scratch.

Latent editing methods exploit the observation that many attributes correspond to directions or subspaces in StyleGAN representations [4, 5]. Attribute Group Editing (AGE) applies this idea to few-shot generation by separating category-relevant class structure from category-irrelevant variations such as color, pose, and texture [6]. Stable Attribute Group Editing (SAGE) improves AGE by estimating a more stable class embedding from few-shot examples and by adaptively selecting compatible attribute directions from related seen classes [7]. However, SAGE still depends on a pSp inversion path, and encoder-based inversion can trade reconstruction fidelity against editability [8, 9].

This study addresses that limitation by integrating the StyleFeatureEditor (SFE) Inverter and Feature Editor into SAGE. Instead of representing an input only through W^+ style codes, FeatureSAGE predicts both a W^+ latent and an intermediate feature tensor, then edits both during synthesis [10]. The research problem is whether this feature-level inversion path can improve detail preservation, edit stability, and distributional quality for novel classes while retaining the SAGE mechanism for class-consistent few-shot attribute group editing.

Specifically, the study asks whether replacing pSp with SFE improves inversion fidelity and image-detail preservation, whether the change affects consistency and diversity for unseen Flowers and Animal Faces classes, and how the proposed framework compares with AGE and SAGE in image quality and edit reliability. These questions are important because few-shot augmentation is useful only when generated samples preserve category identity while still providing realistic non-category variation. A generator that produces visually plausible but class-inconsistent images can reduce downstream value, while a model that preserves the reference too closely may fail to provide useful diversity.

The practical motivation is data-efficient image synthesis for fine-grained domains where collecting many labeled examples is difficult. Animal-face and flower categories contain meaningful intra-class variation in color, texture, orientation, and local structure, but they also require strong preservation of category-specific appearance. This makes them suitable testbeds for evaluating whether a few-shot generator can introduce useful variation without corrupting the class-defining features of the reference image.

The contributions of this paper are as follows: (1) a FeatureSAGE framework that replaces the pSp-based SAGE inversion path with a StyleFeatureEditor Inverter and Feature Editor; (2) a one-shot evaluation protocol on FUNIT Animal Faces and 102 Flowers using the same pretrained StyleGAN2 setting, data splits, and metrics across AGE, SAGE, and FeatureSAGE; and (3) quantitative and qualitative evidence showing that FeatureSAGE lowers FID while keeping LPIPS close to the baselines. The study is limited to StyleGAN2-based few-shot image generation on Animal Faces and Flowers and does not evaluate optimization-based inversion or non-StyleGAN generators.

2. REVIEW OF RELATED LITERATURE

2.1 StyleGAN Latent Editing and Inversion

StyleGAN introduced a style-based generator in which an input latent is transformed into an intermediate W space and injected into synthesis layers, yielding stronger disentanglement and image quality than earlier GAN designs [2]. StyleGAN2 improved this generator by replacing AdaIN with weight modulation and demodulation and by regularizing the latent-to-image mapping, which supports higher fidelity and more stable inversion [3]. These architectures enable latent traversal, where an image can be edited by adding an attribute direction to a latent code. Methods such as InterFaceGAN, GANSpace, and SeFa show that pose, color, expression, and other attributes can often be controlled through approximately linear directions in StyleGAN spaces [4, 5, 11].

The latent representation chosen for editing strongly affects the balance between detail and control. A compact W code tends to stay closer to the generator distribution and supports smoother edits, while W^+ assigns a separate style vector to each synthesis layer and improves reconstruction at the risk of less predictable edits. Feature-space representations can retain still finer spatial information, but they require an editing mechanism that does not simply overwrite details during synthesis. This trade-off is central to FeatureSAGE because few-shot generation needs both stable class-level edits and preservation of local cues such as flower petal structure or animal-face texture.

For real images, latent editing first requires GAN inversion. Encoder-based methods such as pSp, e4e, and ReStyle predict W^+ codes efficiently, while optimization-based methods can improve reconstruction at the cost of speed [8, 12, 13]. The central difficulty is the fidelity-editability tradeoff: high-capacity inversions in W^+ can preserve fine detail, but may move codes away from regions where semantic edits remain reliable [9]. StyleFeatureEditor addresses this limitation by predicting both a W^+ latent and an intermediate feature tensor, then training a Feature Editor to update feature maps during editing [10]. This motivates its integration into SAGE, whose baseline pSp inversion may lose local information that is useful for class-consistent attribute editing.

This distinction matters because a few-shot image generator is not judged only by whether an edited output looks realistic in isolation. It must also preserve the category cues that make the single reference useful. If inversion discards local structure before the edit is applied, the downstream attribute editing stage cannot recover those details reliably. Conversely, if the representation preserves details but becomes too entangled, edits may change category identity. FeatureSAGE is positioned between these needs by retaining SAGE’s disentangled attribute dictionary while using SFE to carry additional feature-level evidence through the decoder.

2.2 Few-Shot Attribute Group Editing

Few-shot image generation seeks to synthesize diverse samples for a novel category from one or a few references. Directly adapting a powerful generator to a very small target set can overfit and reduce diversity, so many

approaches transfer a pretrained generator and regularize adaptation through freezing, contrastive losses, or distance preservation [14–16]. Editing-based methods avoid full retraining by reusing latent directions learned from seen categories.

AGE formulates a category as a combination of category-relevant structure and category-irrelevant attributes, learning a sparse dictionary of transferable directions and applying them to unseen-class embeddings [6]. SAGE improves the stability of AGE by estimating a class embedding using all available few-shot references and by selecting attribute directions from similar seen categories, thereby reducing class-inconsistent edits [7]. TAGE further studies trustworthy attribute group editing using codebooks and prompt-driven semantic modules [17]. These works show that latent attribute dictionaries can support few-shot synthesis without retraining the generator, but their effectiveness still depends on inversion quality and on preserving class identity under multi-attribute changes.

The present work therefore keeps the SAGE factorization and adaptive editing logic rather than introducing a new few-shot generator. Its modification is narrower: the inversion and decoding path is strengthened using SFE so that the latent edit is paired with an edited feature tensor. This scope preserves the original SAGE research line while testing whether the known reconstruction benefits of feature-level inversion translate to one-shot attribute group editing.

The related work also motivates the selected baselines. AGE is the direct attribute-group editing baseline because it demonstrates few-shot generation through category-relevant and category-irrelevant decomposition. SAGE is the stronger baseline because it stabilizes AGE using class embedding mapping and adaptive editing. Comparing FeatureSAGE against both methods isolates whether the proposed inversion and feature-editing path improves a mature editing-based pipeline rather than merely outperforming an older formulation.

3. METHODOLOGY

3.1 Research Design and Data

The study uses a quantitative experimental design that changes one component of SAGE: the GAN inversion path. The baseline SAGE configuration uses pSp, while FeatureSAGE replaces it with the StyleFeatureEditor Inverter and Feature Editor. All comparisons use the same pretrained StyleGAN2 generator, datasets, preprocessing, one-shot setting, random seeds, sample counts, optimizer settings, and metric implementations. Experiments were run on a virtualized NVIDIA GRID V100D-32A environment with 32 GB RAM, Ubuntu 24.04 LTS, PyTorch, Adam optimization, and mixed precision.

Two datasets are used. FUNIT Animal Faces contains 117,484 aligned 512×512 animal-face images across 149 classes, split into 119 seen and 30 unseen classes [18]. The 102 Flowers dataset contains 102 flower categories with 40–258 images per class, split into 85 seen and 17 unseen classes [19]. Images are resized and normalized for the StyleGAN2 and evaluation setting; Animal Faces images are aligned with MTCNN. The term *one-*

shot refers to a single reference image per unseen target class during test-time generation.

The independent variable is the inversion and editing path: pSp for SAGE and SFE plus Feature Editor for FeatureSAGE. Dependent variables are reconstruction and generation quality as measured by FID and LPIPS. Controlled variables include the pretrained generator, seen/unseen class splits, preprocessing, edit sampling procedure, training iterations, optimizer configuration, and evaluation implementation. This control is necessary because the reported differences are intended to isolate the effect of feature-level inversion and editing rather than changes in the generator or data protocol.

Figure 1 illustrates the two evaluation domains. The split into seen and unseen categories follows the few-shot generation setting: seen classes provide latent statistics and attribute dictionaries, while unseen classes test whether those learned directions can transfer from a single reference. No additional augmentation beyond centering, alignment where applicable, resizing, normalization, and simple flipping is introduced, since the purpose is to measure the effect of the inversion and feature-editing path rather than the effect of a data-expansion pipeline.

3.2 FeatureSAGE Architecture

Figure 2 summarizes the proposed pipeline. SAGE represents each category using a class embedding and edits category-irrelevant directions in W^+ . For a class C with N_C latent codes, the category-relevant embedding is

$$e^C = \frac{1}{N_C} \sum_{i=1}^{N_C} w_i^C. \quad (1)$$

In the baseline, pSp predicts $w_i^{C_s} \in W^+$ for seen-category samples. FeatureSAGE instead uses the SFE inverter

$$I(X_i^{C_s}) = (w_{i,\text{sfe}}^{C_s}, F_{k,i}^{C_s}), \quad (2)$$

where $F_{k,i}^{C_s}$ is the intermediate feature tensor injected at generator layer 9. This representation preserves the layer-wise editability of W^+ while retaining feature-level detail for the decoder.

FeatureSAGE keeps the SAGE attribute factorization mechanism. For a seen sample, the category-irrelevant residual is

$$\Delta W_i^{ir} = W_i^{C_s} - e^{C_s}, \quad \widehat{\Delta W_i^{ir}} = A n_i, \quad (3)$$

where A is the global dictionary of category-irrelevant directions and n_i is a sparse code. SVD is applied to seen-class embeddings to obtain the category-relevant dictionary B_f . For an unseen class C_u , SAGE estimates

$$\hat{e}^{C_u} = \frac{1}{N_u} \sum_{i=1}^{N_u} B_f B_f^T W_i^{C_u}. \quad (4)$$

Adaptive editing then selects directions from seen categories closest to C_u , samples a sparse edit code $\tilde{n}_j$, and synthesizes the output as

$$X_j^{C_u} = G(\hat{e}^{C_u} + \alpha A_{C_u} \tilde{n}_j, H(F_{k,i}, \alpha A_{C_u} \tilde{n}_j)). \quad (5)$$

Here α controls edit strength, A_{C_u} is the adaptive category-irrelevant dictionary, and $H(\cdot)$ edits the intermediate feature tensor before generator synthesis.

The architectural change is deliberately localized. Attribute factorization, class embedding mapping, and adaptive editing remain aligned with SAGE so that the comparison focuses on inversion and decoding. In the pSp-based baseline, the reconstruction detail available for later editing is limited by the predicted W^+ code. In FeatureSAGE, the SFE inverter adds $F_{k,i}$, allowing the decoder to carry local feature evidence through the edit. This is especially relevant for fine-grained regions such as petal boundaries, fur texture, facial contours, and color transitions, where latent-only edits may preserve the general class while weakening local realism.

The module flow in Fig. 3 has four roles. Attribute factorization learns a sparse dictionary of category-irrelevant directions from seen categories. Class embedding mapping filters unseen-category latents through category-relevant directions so that one-shot references are not treated as full class distributions. Adaptive editing selects directions associated with similar seen categories and samples them to generate non-category variation. The Feature Editing Decoder receives both the edited latent and the SFE feature tensor, computes the feature displacement, and injects the edited feature map into StyleGAN2 so that the output follows the semantic edit while retaining local detail.

This design also preserves methodological reproducibility. The class embedding in Equation (1) defines the category-relevant center, Equation (3) defines the residual used for dictionary learning, Equation (4) defines one-shot class estimation for unseen categories, and Equation (5) defines the final synthesis step. The same sequence is applied to AGE, SAGE, and FeatureSAGE where applicable; the proposed model differs only in the feature-aware representation and decoder.

3.3 Training and Evaluation Protocol

Training follows two phases. First, the attribute factorization block is trained with the SAGE objective

$$L_{\text{af}} = \|G(e^{C_s} + A n_i) - x_i^{C_s}\|_2 + \lambda_1 \|B^T A\|_F^2 + \lambda_2 \|\sigma(\theta_0 n_i - \theta_1)\|_1. \quad (6)$$

The terms respectively preserve reconstruction, encourage orthogonality between category-relevant and category-irrelevant directions, and enforce sparse activation. The optimizer is Adam with learning rate 1×10^{-4} , $\lambda_1 = 5 \times 10^{-4}$, $\lambda_2 = 5 \times 10^{-3}$, $\theta_0 = 5 \times 10^{-1}$, and $\theta_1 = -1$ [7].

Second, the Feature Editor H is trained on SAGE-style editing pairs and a zero-displacement inversion task. Its objective is

$$L_{\text{fe}} = (L_2 + \lambda_{\text{lpi}} L_{\text{lpi}})_{\text{edit}} + (L_2 + \lambda_{\text{lpi}} L_{\text{lpi}})_{\text{inv}}. \quad (7)$$

The Feature Editor is trained for 15,000 iterations using Adam at 1×10^{-4} , $\lambda_{l2} = 10$, $\lambda_{\text{lpi}} = 0.8$, an additional LPIPS scaling factor of 0.5, mixed precision, and dictionary length 100.

During adaptive editing, residual directions are projected back into sparse-code space as $\hat{n}_i = A^\dagger \Delta W_i^{ir}$. The



Fig. 1. Representative samples from the Animal Faces and 102 Flowers datasets used for few-shot evaluation.

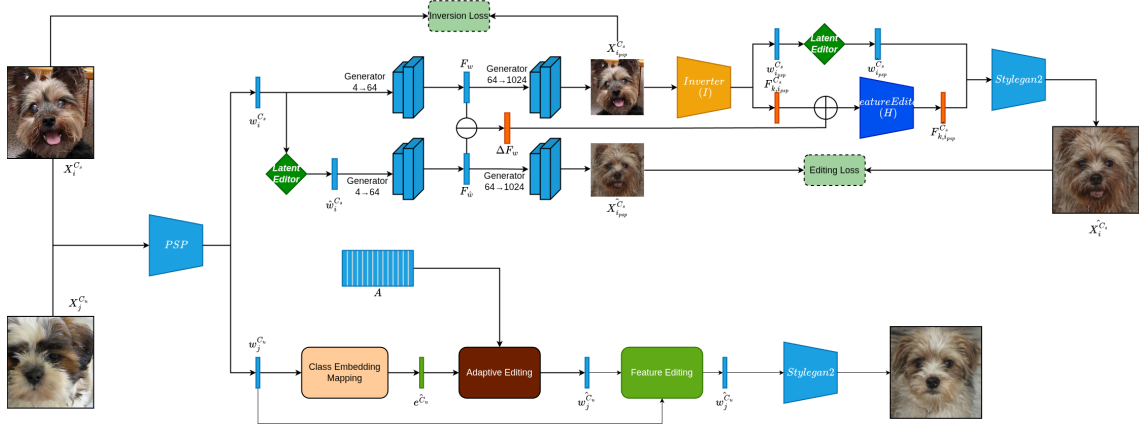


Fig. 2. FeatureSAGE integrates the StyleFeatureEditor Inverter and Feature Editor into the SAGE attribute group editing pipeline.

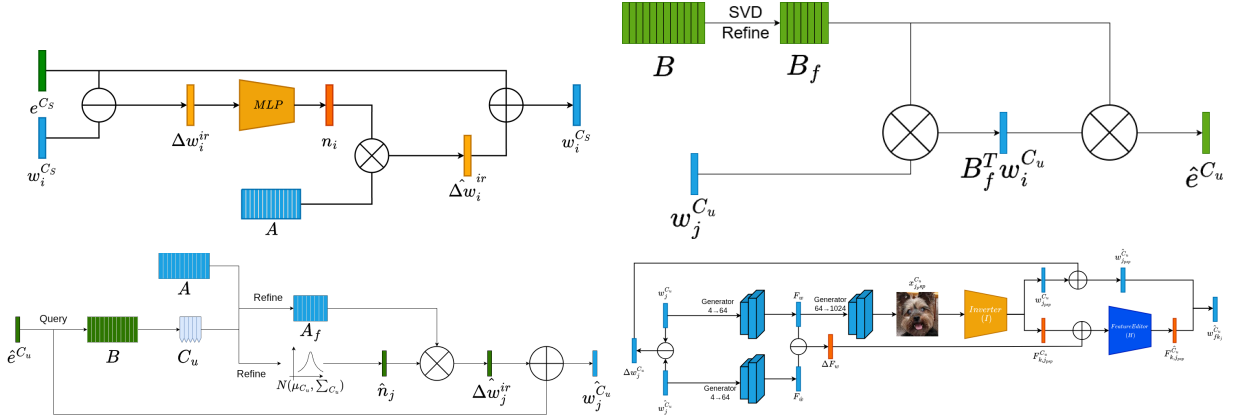


Fig. 3. Core modules retained or adapted in FeatureSAGE: attribute factorization, class embedding mapping, adaptive editing, and the feature editing decoder.

model averages absolute sparse activations across the t_C most similar seen categories and selects the top t_A directions per W^+ layer to form A_{C_u} . For the decoder, the feature displacement is computed from the original and edited generator features,

$$\Delta F_w = F_w - F_{\hat{w}}, \quad (8)$$

and $H(F_{k,i}, \Delta F_w)$ produces the injected feature tensor. This step is the architectural difference from SAGE: SAGE edits only the latent code, while FeatureSAGE edits the latent and the feature representation jointly.

AGE, SAGE, and FeatureSAGE are evaluated at 1024×1024 resolution using FID and LPIPS. FID measures distributional alignment from Inception-v3 features [20]:

$$\text{FID} = \|\mu_r - \mu_g\|_2^2 + \text{Tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2}). \quad (9)$$

Lower FID indicates better alignment between generated and real feature distributions. LPIPS is reported as a per-

ceptual variation measure among generated outputs, with the same implementation used for all methods [21].

The metrics are interpreted jointly. FID alone can improve if generated samples become visually cleaner but less varied, while LPIPS alone can be high for outputs that vary but drift away from the intended category. The desired result is therefore a lower FID with LPIPS remaining close to the baselines. This is the evaluation criterion used in the results section when judging whether FeatureSAGE improves quality without removing useful output variation.

LPIPS compares deep feature activations rather than raw pixels and is computed as

$$\text{LPIPS}(x, x') = \sum_l \frac{1}{H_l W_l} \sum_{h,w} \|w_l \odot d_{hw}^l\|_2^2. \quad (10)$$

Here $d_{hw}^l = \hat{y}_{hw}^l(x) - \hat{y}_{hw}^l(x')$, $\hat{y}_{hw}^l$ denotes normalized features at layer l , w_l is the learned channel weighting, and H_l, W_l are feature-map dimensions [21]. In this pa-

Table 1. Quantitative comparison at 1024×1024 resolution. Lower FID is better; LPIPS is reported as generated-sample perceptual variation.

Model	Flowers		Animal	
	FID	LPIPS	FID	LPIPS
AGE	61.57	0.4108	62.13	0.5033
SAGE	54.17	0.4528	60.92	0.4867
FeatureSAGE	45.60	0.4209	49.60	0.4811

per, LPIPS is used to describe perceptual variation among generated samples rather than paired reconstruction error. This distinction is important because the proposed method is expected to reduce FID without making generated samples visually identical.

The baselines are evaluated under the same one-shot query setting. AGE starts from a class embedding and samples category-irrelevant attribute directions from its learned dictionary [6]. SAGE improves this process by estimating a stable class embedding and selecting directions from related seen classes [7]. FeatureSAGE follows the same editing protocol as SAGE but substitutes the inversion and decoder path with SFE-based latent and feature editing. Therefore, the comparison measures whether feature-level detail preservation improves an otherwise controlled SAGE pipeline.

All results are reported without introducing new post-processing or additional training data. The generated images used for FID and LPIPS are produced by the corresponding model configuration and evaluated against the same dataset split. This avoids confounding the effect of FeatureSAGE with changes in sample filtering, manual selection, or evaluation subsets.

4. RESULTS AND DISCUSSION

Table 1 compares AGE, SAGE, and FeatureSAGE under the one-shot protocol. FeatureSAGE obtains the lowest FID on both datasets, indicating improved distributional alignment after replacing the pSp-based SAGE inversion path with feature-level inversion and editing.

On 102 Flowers, FeatureSAGE reaches an FID of 45.60, improving over SAGE by 15.8% and AGE by 25.9%. Its LPIPS of 0.4209 is lower than SAGE by 7.0% and higher than AGE by 2.5%, showing that the FID gain does not collapse the outputs into a narrow perceptual range. On Animal Faces, FeatureSAGE reaches an FID of 49.60, improving over SAGE by 18.6% and AGE by 20.2%. Its LPIPS of 0.4811 remains close to SAGE (0.4867) and AGE (0.5033), within 4.4% of both baselines.

The different LPIPS directions across datasets are consistent with the qualitative behavior. On Flowers, FeatureSAGE reduces some of the broader variation produced by SAGE while improving realism and petal coherence. On Animal Faces, the LPIPS reduction relative to AGE is small, but the samples show less structural drift. Thus, the main measured gain is not higher diversity; it is better distributional alignment while keeping generated outputs perceptually varied enough to remain useful for few-shot augmentation.

The improvement is also consistent across both domains

rather than being isolated to a single dataset. Flowers provide a fine-grained object domain where color and shape variation can easily change class appearance, while Animal Faces test aligned facial structure and texture. FeatureSAGE improves FID in both settings, suggesting that the SFE-based path benefits the SAGE editing process in more than one structured visual domain. The retained LPIPS range further indicates that the model does not simply average away variation to obtain cleaner images.

These results address the four research questions stated in the introduction. First, the lower FID values support the claim that SFE-based inversion and feature editing improve the generated distribution relative to the pSp-based SAGE path. Second, the comparable LPIPS values show that the change does not remove perceptual variation under the one-shot setting. Third, the qualitative samples indicate more reliable structure and fewer unstable local edits, particularly in flower edges and animal-face regions. Fourth, the comparison with AGE and SAGE shows that the improvement is measured against editing-based few-shot generators rather than against unrelated synthesis methods.

The qualitative comparison in Fig. 4 supports the quantitative trend. On Flowers, FeatureSAGE better preserves sharp petal edges, contrast, and color coherence than the baselines, although some input-specific color details such as combined pistil colors are not always reproduced exactly. On Animal Faces, all methods preserve broad color and texture, but FeatureSAGE produces more stable shape, eye structure, fur shading, and pose consistency. AGE and SAGE sometimes show greater orientation diversity, while FeatureSAGE favors structural coherence and realism.

These observations also identify the remaining failure modes. FeatureSAGE can still miss fine input-specific color patterns, and its stronger consistency can reduce extreme pose variation relative to AGE. This means the model is better interpreted as improving stable, class-preserving variation rather than maximizing every possible form of diversity. The main system limitations are the dependence on a pretrained StyleGAN2 generator and the added computational cost of the SFE inverter, feature extraction, feature editing, and feature injection steps. Thus, FeatureSAGE improves fidelity and stability within the tested StyleGAN2 domains, but scaling to other generators or faster inference remains future work.

The findings should therefore be interpreted within the stated scope. The experiments do not claim that FeatureSAGE solves few-shot generation for arbitrary domains, nor do they evaluate diffusion or transformer-based generators. Instead, the results show that within a StyleGAN2 and SAGE setting, replacing latent-only inversion with joint latent-feature editing improves distributional alignment while preserving a usable degree of perceptual variation.

5. CONCLUSION

This paper presented FeatureSAGE, a SAGE-based few-shot image generation framework that integrates the StyleFeatureEditor Inverter and Feature Editor to improve latent inversion and feature-level detail preserva-

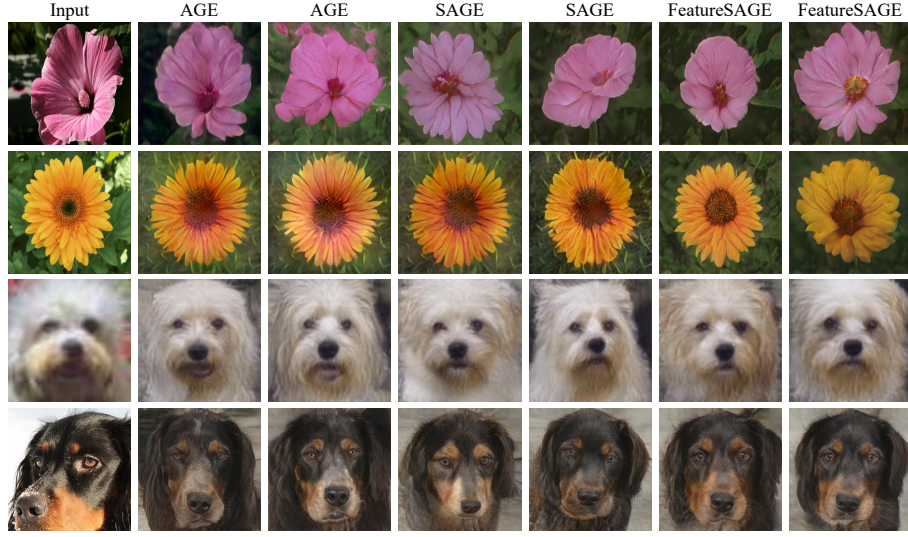


Fig. 4. Qualitative comparison on Flowers (top two rows) and Animal Faces (bottom two rows).

tion. The method keeps the pretrained StyleGAN2 generator and the SAGE attribute group editing mechanism, but represents each image using both W^+ latents and intermediate feature tensors.

The results answer the study objectives affirmatively. At 1024×1024 resolution, FeatureSAGE achieved an FID of 45.60 and LPIPS of 0.4209 on 102 Flowers, improving FID by 15.8% over SAGE and 25.9% over AGE. On Animal Faces, it achieved an FID of 49.60 and LPIPS of 0.4811, improving FID by 18.6% over SAGE and 20.2% over AGE. The LPIPS values remained close to the baselines, indicating that the FID improvements were obtained while retaining comparable perceptual variation.

Overall, FeatureSAGE strengthens one-shot attribute group editing by combining latent-space control with feature-level detail preservation. Its limitations are the need for a suitable pretrained StyleGAN2 generator and the additional computation introduced by the SFE-based inversion and editing path. Future work should examine faster inference, alternative generator backbones, and multi-level feature editing while preserving class-consistent few-shot synthesis.

For practical use, the results suggest that feature-level editing is most valuable when the target task rewards coherent class-preserving samples rather than maximal pose or appearance spread. The framework therefore fits data augmentation scenarios where synthetic images must remain visually close enough to the target category to support downstream learning.

The final paper therefore preserves the original research narrative: few-shot synthesis remains limited by the tension between reconstruction fidelity, editability, and class consistency; SAGE supplies a stable attribute group editing framework; and FeatureSAGE improves that framework by replacing latent-only inversion with joint latent and feature editing.

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